

IPv6

This IP Address Format will be gone soon. Enter IPv6

2001:0db8:3c4d:0015:0000:

Ram Mohan Rao

If you're reading this, you know what an IP (Internet Protocol) is—it looks something like 202.144.98.72, and all Internet hosts have one. Now, what we use today is called IPv4—version 4; the not-so-distant future network will use IPv6, or IPng (the “ng” is for “Next Generation”).

Address Space

The first and most oft-cited reason for the move to IPv6—the evolutionary successor to IPv4—is the address space. IPv4 allows for about four billion unique addresses, which seems enough—after all, there are only about six billion people. But there are at least two reasons why we'll be running out of addresses. First, IPv4 addresses are classified as Class A, Class B, and so on. Think of IBM, which has been assigned the Class A range of 9.0.0.0 to 9.255.255.255. This is almost 17 million addresses—most of them unused, of course. Simply telling IBM to take a smaller range just won't do it! Second, and more interesting, is the fact that more and more devices—not just computers—will be connected to the information network of the future: your mobile phone (easy to believe), your smart fridge (a little futuristic), and even your alarm clock (sounds way out, but you better believe it)! To accommodate all these will require something much more than four billion, and that's one major *raison d'être* for IPv6.

Why Not NAT?

You might have heard of NAT (Network Address Translation). An office, for example, might have a LAN, where 40 individual computers show up as just one IP to the external world—the NAT router takes care of what traffic from the outside should be routed to which computer on the inside. Now, this may look like a perfect solution, since 39 IPs have been conserved. There are several reasons cited for why this is not the ideal solution, amongst them the fact that hosts on the outside with “real” addresses can't initiate communication with the “NATted” computers. Also, direct, peer-to-peer communication is not possible if a network is NATted, because of the router.

What It Looks Like

There are different types of IPv6 addresses, but not to get into the details, a typical address looks like FECC:B672:391C:2322:CD51:AAEE:3DEC:0921 (instead of like 202.144.98.72).

This is a stringing of eight 16-bit hexadecimal values, and means a 128-bit address space—which in turn means 3.4×10^{38} addresses—you don't need to imagine that number; it's practically infinite!

If an address has long sub-strings of all zeroes,

the sub-strings can be abbreviated by a double colon. In addition, up to three leading zeroes per four hex values can be omitted. Taking both these together, FECC::1 corresponds to FECC:0000:0000:0000:0000:0000:0000:0001.

Routing Tables

Routing tables are what enable your Internet packets to reach their destination—they contain information about where a packet should go next en route to its destination. With IPv4, the Internet backbone routers—which control Internet traffic at the top level—contain routing tables that are already very large, and are growing. This means inefficiency, and further growth will hamper their very functioning.

Now, IPv6 has been designed so that Internet backbone routers will need to have much smaller routing tables. The tables, instead of including every possible route, need only include routes to those routers that are directly connected to them. How that works is beyond our scope here, but suffice it to say that IPv6 solves “the exploding routing table problem” to a large extent.

Other Goodies

There are several other advantages of IPv6 that justify a worldwide switchover. For example, it Quality of Service (QoS) is inbuilt in IPv6; this, while not essential, is a good thing for VoIP and multimedia, for example. It also allows for prioritisation of data—time-sensitive streams such as videoconferencing data can be assigned a higher priority than, say, Web browser requests.

Then, in the realm of security, consider IPSec. Short for “IP Security,” it is a set of protocols to support secure exchange of packets. IPSec is widely used in the implementation of Virtual Private Networks (VPNs). IPSec is optional in IPv4; in IPv6, it's embedded in the headers. Setting up a VPN through IPv4 requires confirmation that the other user also supports IPSec; IPv6 will eliminate this requirement.

IPv6 brings with it new functions that simplify the configuration and management of the addresses on a network, which are typically labour-intensive. Several tasks typically performed by a system administrator are automated. For example, the auto-configuration feature in IPv6 can automatically configure router and interface addresses.

There are many more good things about IPv6, and even more sites from where to get information; for a start, you could try www.ipv6.org.

IPv6 is already being deployed in some areas, especially in Asia. Worldwide deployment is not a whether, but just a “when”. As for the “How” of the matter, that's a story for a different time! ■

ram_mohan@thinkdigit.com